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Professional software for 3D printing

During the initial days of the evolution of 3D printing, either open source software for DIY printers or proprietary codes by commercial 3D printers were widely in use. Hence, there was always a dearth and huge demand of professional high-end 3D printing and additive manufacturing software with requisite features, such as model repair, proficient slicing, 2D and 3D nesting of parts in a build volume, support generation, thumbnail rendering, texture and colour support, job optimisation and job queuing. Now the industry is exploring professional software that allows you to make late design changes along with reduced risk of failed builds and wasted material, thereby preventing machine errors and enhancing overall process reliability and efficiency. The wishlist of features is ever demanding and growing, like Cloud services, efficient connect and porting from Microsoft Kinect, or other sensors and 3D scanners. Currently, netfabb Pro is one of the only popular 3D printing software, which is not only used for Fabbster printer but other commercial printers, like Voxeljet, Rapidshape, and Ultimaker, among others. In 2015, we can anticipate new software entrants with interesting features and improvised functionalities.

—Contributed by **Vinay Sharma**, director-technical, ni logic Pvt Ltd (ni2designs)

Price trends

According to Rodagi, a notable change in this arena is the cost and awareness factor. He says, "About five years ago, costs were exorbitant, which was one of the barriers against the use of this technology. Prices of 3D printers will continue to become more user-friendly in the near future." There are low-cost printers that start from ₹ 50,000 (or even lesser), which are best used for DIY projects, whereas a professional Stratasys 3D printer starts from ₹ 650,000. He adds, "3D printers that can print metal are quite expensive today, but their prices should come down in a couple of years at the most."

Although the cost of printers has been decreasing drastically over the past two years (making desktop printers much more affordable), current bottlenecks still exist in print times and the cost of the filament itself, informs Pai. However, he says, "Improved FDM printer designs and the advent of low-cost SLA printing have the potential to greatly improve desktop print speeds. The filament and raw material costs will also see a significant drop as 3D printing gains mainstream adoption."

With the Chinese entering the market with force, prices have continued to move south. Shishodia says, "Prices of entry-level desktop 3D printers start from ₹ 50,000 and may go up to ₹ 60,000. Good DIY kits retail from ₹ 35,000 to ₹ 40,000." As more and more players enter the market, prices are further expected to fall.

Depending on technology and requirements, one could get an FDM/FFF machine for as low as ₹ 20,000 to an average higher price of anywhere between ₹ 400,000 and ₹ 500,000, informs Reddy. He says, "An SLA/direct light processing (DLP) machine costs anywhere between ₹ 400,000 and one million. Anything branded from big names, like 3D systems or Stratasys, is competitively very expensive."

"These advancements could also give rise to a number of moral issues, which will need to be resolved as they arise. Either way, the next few years promise to be interesting."

3D hubs and print shops can be expected to open everywhere, from primary schools to homes to industrial houses. "3D printers will become extremely versatile machines to be

used in all walks of life. Further, networked 3D printers may enable users to detect and better utilise spare capacity," says Datla.

3D printers are the next big thing

The impact of 3D printing on technology and everyday life will be unprecedented. "The remarkable possibility of widespread domestic use of this technology has tremendous potential to change the way goods are obtained, designed and innovated," says Datla.

Technically, a patent expired in January 2013, which unlocked a high-resolution 3D-printing technique using light/laser, informs Reddy. (A patent is usually valid for 20 years and, after expiry, it becomes open source and free for access to anyone.) Many patents in 3D printing have already expired, or will soon expire, and this will enable even faster development and proliferation of 3D printers.

3D printers are the next big thing in the field of technology, feels Shishodia. He says, "The day is not far when we will all have a 3D printer on our desktops, as we have normal printers." Reddy too feels that a 3D printer would be in every home, like a computer, within the next ten years.

Aakash shares, "General acceptance of 3D printing, along with its limitations, has found a place in the minds of consumers." It can be comparable to the acceptance and popularity of dot-matrix printers. He adds, "Digital manufacturing will only get bigger in the future." ●